

Cortical brain potentials in response to lower limb proprioceptive stimuli in young adults with probable developmental coordination disorder

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ABSTRACT

Background: Proprioceptive deficits have been shown to underlie motor problems in individuals with a probable developmental coordination disorder (pDCD). Behavioral studies have employed response times to passive limb movement to evaluate proprioceptive function in individuals with pDCD. However, the underlying neural mechanisms involved in the cortical processing of proprioceptive input and its corresponding motor response are unclear. To address this issue, this study aims to investigate neuropsychological and neurophysiological performances using event-related potentials (ERP) on proprioceptive-motor tasks in young adults with pDCD.

Methods: From a total of 149 young adults screened using the Bruininks-Oseretsky Test of Motor Proficiency 2nd Edition Complete Form (BOT-2), 12 individuals with pDCD were identified (mean age $\pm$ SD: 20.50 $\pm$ 1.08 years) along with 12 age- and sex-matched controls (mean age $\pm$ SD: 20.75 $\pm$ 1.05 years). Participants placed their dominant foot on a passive ankle motion apparatus that plantarflexed the ankle at a constant velocity of 22°/s for a total of 75 trials in each proprioceptive condition. With vision occluded, participants had to press the trigger button held by the dominant hand when they sensed the passive motion of the ankle (voluntary response, VR), or purely receive the movement without a voluntary response (non-voluntary response, NVR). Behavioral performances [i.e., mean movement detection time (MDT_{mean}), the standard deviation of the movement detection time (MDT_{SD})] and ERP indices (i.e., N1, P3 amplitude, and latency) related to ankle kinesthetic stimuli were obtained to determine the proprioceptive-motor function.

Results: The results showed that young adults with pDCD exhibited longer MDT_{mean} ($p < 0.001$) and MDT_{SD} ($p = 0.002$) compared to their controls. Electrophysiological indices measured at frontal and central electrode sites, showed that young adults with pDCD exhibited significantly smaller N1 ($p = 0.019$) and P3 amplitudes ($p = 0.032$) during VR and NVR conditions. Notably, correlation analysis revealed a significant negative relationship between MDT_{mean} and N1 ($r = 0.62, p < 0.001$) and P3 amplitudes ($r = -0.55, p = 0.005$) in the VR condition in young adults with and without pDCD.

Conclusions: This study sheds light on the central brain mechanisms underlying proprioceptive-motor deficits in young adults with pDCD. The combined analysis of behavioral and ERP data suggests that longer MDT_{mean} and larger MDT_{SD} in young adults with pDCD are associated with weaker proprioceptive afferent inflow shown by decreased N1 amplitude to the frontal and parietal cortices. Such degraded proprioceptive signals are followed by reduced P3 amplitude, suggesting that young adults with pDCD allocate fewer neural resources to modulate motor processes with regard to proprioceptive stimuli. These findings contribute to a better understanding of the neurophysiological basis of proprioceptive-motor deficits in pDCD and may inform the development of targeted sensorimotor interventions.

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1. Introduction

Developmental coordination disorder (DCD) is a neurodevelopmental condition characterized by impaired motor coordination that significantly interferes with daily activities and academic achievement (Inder & Sullivan, 2005). Individuals with DCD exhibit difficulties in motor learning, planning, and execution, which cannot be attributed to intellectual disability, visual impairment, or other neurological conditions affecting movement (Wilson et al., 2017). DCD affects approximately 5–6 % of school-aged children, with many individuals continuing to experience these difficulties into adolescence and adulthood (Hyde et al., 2018; Kirby et al., 2013; Tal-Saban et al., 2014). The population of individuals with DCD is heterogeneous, as evidenced by abnormalities in neural networks involving frontal and parietal regions, basal ganglia, and the cerebellum, associated with neurocognitive and neuromotor functions in populations with DCD (Biotteau et al., 2016).

One crucial aspect of motor control and learning that may be impaired in DCD is proprioception (Chen et al., 2020; Tsay et al., 2021; Tseng et al., 2022; Tseng et al., 2018, 2019). Proprioception, the sense of body position and movement derived from receptors in muscles, tendons, and joints, is a crucial aspect of motor control and learning that may be impaired in DCD (Proske & Gandevia, 2012; Tsay et al., 2021). This sensory system plays a vital role in how individuals perceive and interact with their environment, bridging the gap between body and space. Accurate proprioceptive feedback is essential for the development and execution of coordinated movements (Ostry et al., 2010; Tsay et al., 2021). Proprioception functions at both conscious and unconscious levels, with conscious proprioception contributing to muscle activation during voluntary movements and unconscious proprioception facilitating subconscious monitoring of body positioning (Holst-Wolf et al., 2016; Proske & Gandevia, 2012). Disruptions in proprioceptive processing can result in altered body-space interactions, manifesting as motor coordination challenges, reduced movement accuracy, and difficulties in navigating and performing activities in daily environments (Hsu & Tseng, 2024; Konczak et al., 2009; Tseng et al., 2023).

Proprioceptive deficits have been observed in individuals with DCD through various behavioral studies, suggesting a potential link between impaired proprioception and the motor difficulties of DCD. These deficits are particularly evident in lower limb function, affecting gait and posture (Chen et al., 2020; Tseng et al., 2022; Tseng et al., 2018; Tseng et al., 2023). For instance, Chen et al. (2020) found larger position sense errors and variability in the knee and ankle of individuals with DCD compared to controls, with the ankle being more affected, and these deficits were correlated with poorer balance. Similarly, Tseng et al. (2023) observed prolonged ankle motion detection times in individuals with DCD, and decreased ankle motion sense sensitivity was associated with poorer lower limb motor function. These findings collectively suggest that proprioceptive deficits in the lower limbs are common in DCD and likely contribute significantly to the motor coordination difficulties and poorer lower extremity performance observed in this population.

While behavioral findings provide valuable insights into proprioceptive impairments in DCD, they offer limited understanding of the neural mechanisms underlying these deficits. Behavioral measures of proprioception, such as joint position errors and motion detection times can be confounded by cognitive and motor control factors and do not directly reveal the cortical processing of proprioceptive input and motor output (Goble et al., 2009; Riemann & Lephart, 2002). To address this limitation and elucidate the neural underpinnings of proprioceptive deficits in DCD, neurophysiological techniques that directly assess the cortical processing of proprioceptive information and its relation to motor response are necessary.

Event-related potentials (ERP) provide a direct approach to observing neural responses associated with sensory and motor processing with high temporal resolution (Luck, 2014). In the context of proprioceptive-motor processing, two ERP components are of particular

interest: the N1 and P3 (Toledo et al., 2016). The N1 component of the somatosensory ERP, occurring around 100–150 ms after stimulus onset, reflects early cortical processing of proprioceptive afferent input from peripheral receptors (Luck, 2014; Toledo, Manzano, et al., 2016). Importantly, Toledo et al. (2016) demonstrated that individuals exhibiting reduced N1 amplitude and delayed N1 latency in response to passive ankle movements showed a significant correlation with prolonged reaction time in detecting these movements. This finding suggests that deficits in the initial cortical processing of proprioceptive signals may underlie slowed sensorimotor responses (Toledo et al., 2016), which is particularly relevant to the proprioceptive deficits observed in individuals with DCD.

In addition to the N1, the P3 component, typically observed around 300–600 ms after stimulus onset, is widely believed to represent sensorimotor-related stimulus evaluation and its attentional processes (Polich, 2007). Notably, Tsai et al. (2009) found that children with DCD exhibited smaller P3 amplitudes, especially in tasks requiring a motor response, and this was followed by longer target-N1 latencies in response to visual stimuli. These findings suggest slower early sensory processing associated with target identification and stimulus evaluation, which are crucial for efficient motor execution. Given the proprioceptive deficits and motor difficulties observed in individuals with DCD, it is plausible that similar alterations in the N1 and P3 components may be present during proprioceptive processing and subsequent motor response.

The aim of this study was to investigate cortical brain responses during a proprioceptive-motor task in young adults with and without a probable developmental coordination disorder (pDCD). Specifically, we compared behavioral performances (response times to detect passive ankle movements) and ERP indices of early somatosensory processing (N1) and higher-order sensorimotor processing related to stimulus evaluation (P3) elicited by passive ankle movement in young adults with and without pDCD. To further elucidate the relationship between behavioral proprioceptive performances and its cortical correlates, we examined how response times in the proprioceptive-motor task correlated with N1 and P3 amplitudes and latencies. Through this approach, we sought to investigate whether deficits in the cortical processing of proprioceptive signals, as reflected by ERP measures, were associated with slowed sensorimotor responses at the behavioral level in young adults with pDCD.

We hypothesized that young adults with pDCD would exhibit: a) prolonged response times and larger response variability in detecting passive ankle movements, indicating reduced proprioceptive sensitivity; b) reduced N1 and P3 amplitudes, reflecting impaired early cortical processing of proprioceptive input and deficits in higher-order cognitive processing related to proprioceptive stimulus evaluation; c) significant correlations between behavioral performance (i.e., motion detection time, motion detection variability) and ERP measures (N1 and P3 components), suggesting a tight link between slowed sensorimotor responses and deficits in the cortical processing of proprioceptive signals. By examining ERP components in response to proprioceptive stimuli and their correlation with behavioral measures, we aimed to offer important insights into the cortical mechanisms behind proprioceptive deficits in pDCD and their impact on observable motor performance.

2. Method

2.1. Participants

Young adults were screened from a pool of 149 individuals using the Bruininks-Oseretsky Test of Motor Proficiency 2nd Edition (BOT-2, long form) and the Adult Developmental Coordination Disorder/Dyspraxia Checklist (ADC). To meet the DSM-V criteria for DCD, young adults in the pDCD group had to satisfy the following criteria: 1) BOT-2 total standard scores less than 40, which equals one standard deviation below the mean (criterion A) (Tseng et al., 2022); 2) ADC total scores (part A,

B, and C) higher than 25, and part A scores (childhood) greater than 6 (criterion B and criterion C) (Hyde et al., 2018). Additionally, participants with self-reported neurological conditions (e.g., cerebral palsy, muscular atrophy), behavioral problems (e.g., attention deficit hyperactivity disorder, autism spectrum disorder), current upper limb injuries, or an IQ below 85 were excluded from the study (criterion D). We verified the normal IQ of participants based on their status as college students who had passed the college entrance examination in Taiwan (Tseng et al., 2022).

Through this screening process, we identified 12 young adults with pDCD (mean age $\pm$ SD: 20.5 $\pm$ 1.08 years). We then randomly recruited 12 age- and gender-matched young control participants (M $\pm$ SD: 20.75 $\pm$ 1.05) from the control group (see Table 1). To determine the participants' dominant hand, we utilized the Edinburgh Handedness Inventory (Oldfield, 1971). Foot dominance was assessed by asking participants which leg they would use to kick a ball toward a target (van Melick et al., 2017). This study obtained approval from the Research Ethics Committee at National Tsing Hua University in Taiwan, and all participants gave informed consent prior to data collection.

2.2. Apparatus

To assess the behavioral response to lower limb proprioceptive stimuli, we employed a customized ankle motion apparatus previously used in individuals with motor difficulties (Hsu & Tseng, 2024; Tseng et al., 2023). The ankle motion apparatus comprised a plastic footplate and a control cabinet. The footplate was driven by a servo system consisting of servo drives (MBDLT25SF, Panasonic) and servo motors (HMF042L1U2M, Panasonic) with a precision of 4.29°/s. The signals were seamlessly integrated into the program at a sampling rate of 200 Hz. The device generated movement speeds ranging from 0.01°/s (the slowest) to 30°/s (the fastest), with a 0.01°/s increment. The angular displacement range for ankle plantarflexion was between 0° and 45°.

To capture cortical neural activity concurrent with lower limb proprioceptive stimuli and their corresponding motor responses, electroencephalographic (EEG) recordings were synchronized with the ankle motion apparatus. EEG data were recorded using a SynAmps RT 64-channel amplifier and CURRY8 software package (Neuroscan Inc.)

Table 1

Participant demographics. Mean (SD) of each variable for the probable DCD (pDCD) and control groups.

	pDCD	Control	<i>p</i>
N (male/female)	12 (7/5)	12 (7/5)	
Age (years)	21.25 (1.30)	21.25 (1.50)	
Handedness (right/left)	10/0	10/0	
ADC			
Childhood	12.6 (4.9)	4.0 (6.7)	0.002**
Adulthood	41.5 (14.2)	16.0 (11.6)	< 0.001***
Total ^c	54.1 (18.7)	20.0 (17.0)	0.002**
BOT-2-LF ^a			
Fine manual control	44.43 (10.33)	56.17 (7.04)	0.024*
Manual coordination	36.92 (6.07)	53.50 (7.93)	< 0.001***
Body coordination	38.40 (8.16)	58.10 (8.74)	< 0.001***
Agility and strength	39.00 (4.22)	60.83 (8.93)	< 0.001***
Total motor composite ^b	36.5 (3.94)	61.60 (8.68)	< 0.001***

ADC: Adult Developmental Coordination Disorder/Dyspraxia Checklist.

BOT-2-LF: Bruininks Oseretsky Test of Motor Proficiency 2nd Edition, Long-Form.

* $p < 0.05$, two-tailed *t* tests.

** $p < 0.01$, two-tailed *t* tests.

*** $p < 0.001$, two-tailed *t* tests.

Note: ^a standard score with a mean of 50, standard deviation of 10.

^b sum of the four motor-area composite standard scores.

^c sum of the childhood and adulthood scores.

with a 32-channel Quick Cap C190 (Electrode Positioning Cap, Neuroscan Inc.) designed for the international 10–20 system. Recordings were referenced to averaged mastoids and impedance was maintained below 10 k Ω . Additional electrodes were positioned above and below the left orbit and on the outer canthus of each eye to monitor vertical (VEOG) and horizontal electro-oculographic activity, which was used to identify and remove ocular artifacts from the EEG data, even in the absence of visual stimuli during the proprioceptive tasks. The integration of EEG recordings with the ankle motion apparatus allowed for the simultaneous assessment of neural responses and behavioral reactions to ankle motion stimuli, providing insights into the neural correlates of proprioceptive processing and motor responses.

2.3. Experimental procedure

The experiment was conducted in a sound-shielded room in the EEG laboratory. Participants were comfortably seated on adjustable chairs in front of an ankle apparatus, with their dominant foot resting on the pedal. The ankle apparatus was positioned based on each participant's body measurements to ensure the pedal was initially positioned at a 45° angle relative to the horizontal, with the ankle at a 90° angle (see Fig. 1A). Participants were instructed to keep their tested dominant ankle relaxed throughout the experiment. Surface electromyography was utilized to monitor muscle activity to ensure passive movements and prevent interference with motion detection. Participants wore opaque goggles to eliminate visual cues (see Fig. 1B). EEG was simultaneously recorded during the proprioceptive tasks.

The experimental paradigm consisted of a passive motion detection task adopted from previously employed tasks with both young adults, older adults, and individuals with DCD (Toledo et al., 2016; Tseng et al., 2023). Two conditions were included: The voluntary response (VR) condition, where participants actively responded to proprioceptive stimuli by pressing a button upon detecting the onset of ankle movement, and the non-voluntary response (NVR) condition, where participants received passive ankle movement without any voluntary response (see Fig. 2).

In the VR condition, the ankle apparatus induced foot plantarflexion at 22°/s. Participants pressed a trigger button with their dominant hand upon detecting ankle movement. Each trial began with a brief auditory cue, a beep sound lasting for 0.1 s, initiating passive plantarflexion movement within a randomized interval of 1 to 9 s after the cue. Participants had to press the button as soon as they sensed the ankle motion. After pressing the button, the pedal returned to the starting position. A total of 75 trials were conducted.

In the NVR condition, participants experienced similar ankle motion without the need for a voluntary response. Trials began immediately following an auditory cue, with the pedal moving from 45° to 90° and then returning to the starting position. Again, 75 trials were conducted. Before the actual test, practice trials were administered to ensure participants understood the task, with breaks provided if needed. The two conditions were counterbalanced to eliminate any potential order effects. The entire task for both VR and NVR conditions lasted approximately 20–25 min.

2.4. Data processing

2.4.1. Behavioral measures

Motion detection time (MDT) during the VR condition was calculated. Behavioral data were excluded under the following cases: 1) when participants failed to press the button or pressed it incorrectly (orientation error), or 2) if MDT was less than two standard deviations below the participants' mean MDT (anticipatory error) or greater than two standard deviations above their mean MDT (delay error). Subsequently, two behavioral measures were obtained: mean motion detection time (MDT_{mean}) and its standard deviation (MDT_{SD}) across all trials. MDT_{mean} indicates the speed of detecting and responding to ankle motion, while

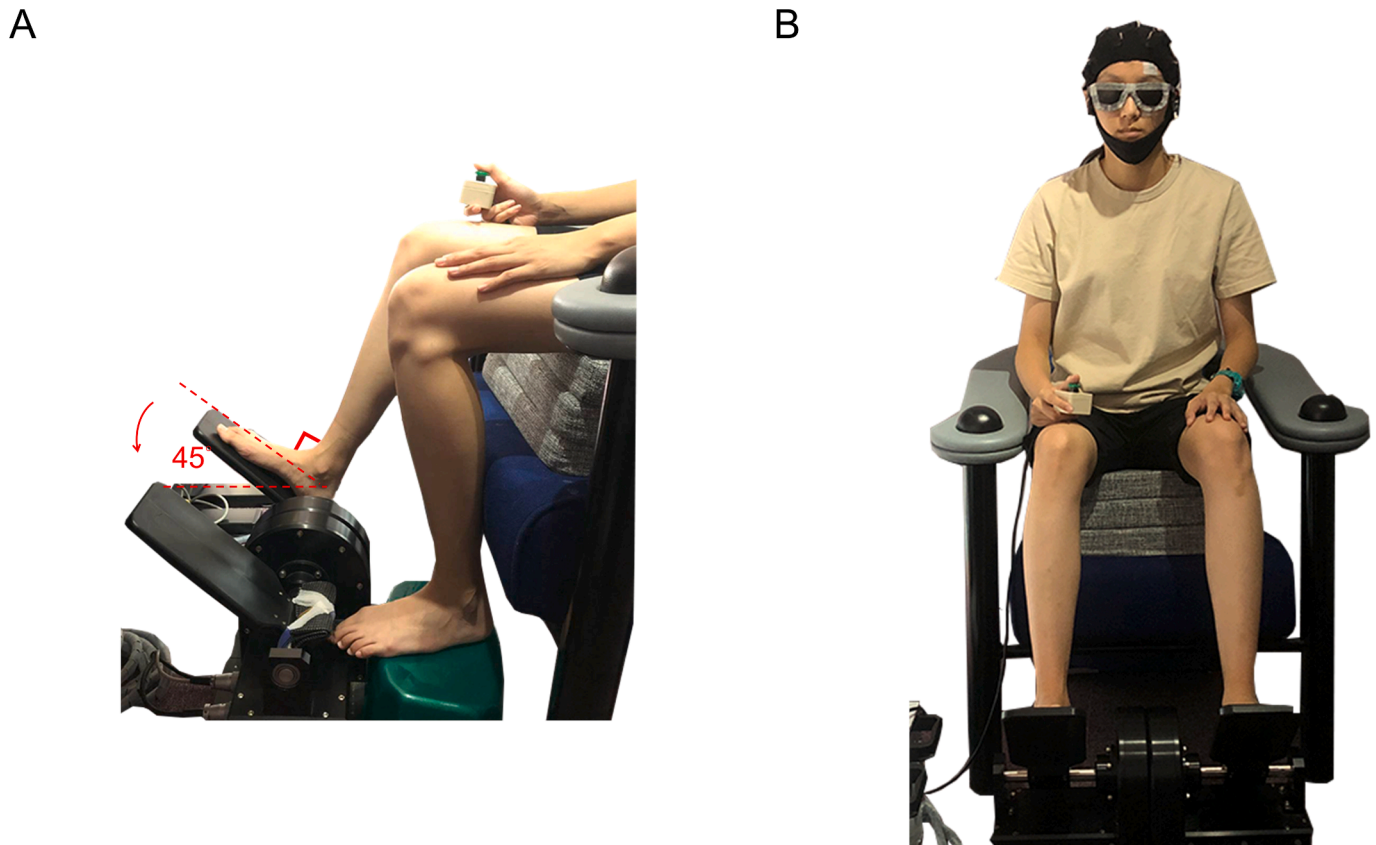


Fig. 1. Experimental setup for the proprioceptive task. (A) Close-up view of the ankle position in the ankle apparatus. The starting ankle angle is set at 45° relative to the horizontal, with plantarflexion movement occurring at a velocity of 22°/s. (B) Full view of the experimental setup showing a participant wearing opaque goggles and an EEG cap while seated comfortably in front of the ankle apparatus for voluntary movement during the proprioceptive task.

MDT_{SD} reflects its response variability.

2.4.2. ERP measures

Offline data processing was conducted using Matlab (R2021a, Mathworks Inc.) and the ERPLAB toolbox (version 4.0.2.3). The epoch file covered the period from −500 ms before stimulus onset to 1000 ms after. Artifact rejection procedures were applied to eliminate epochs with large amplitude artifacts, excluding those with amplitudes exceeding −100 μ V or 100 μ V. Subsequently, outlier data within epochs were removed, and the remaining data were baseline-corrected using the average amplitude of the −200 ms pre-stimulus period. Individual averages were obtained by averaging data from 75 trials for each subject, followed by group averaging for each experimental group. EEG data were extracted from midline channels (Fz, FCz, and Cz) located over the sensorimotor cortex, which is known for processing somatosensory input from the lower limbs and motoric stimulus evaluation (Toledo, Manzano, et al., 2016; Tsai et al., 2012). These channels were selected for their robust signals related to ERP components associated with ankle proprioceptive-motor tasks and stimulus evaluation. Peak detection algorithms were employed to identify and locate brainwave peak values, and marker reports were generated for peak value output. Finally, ERP graphs were plotted to visualize the EEG data. The N1 component, reflecting early sensory discriminative processing, was identified as its peak negative deflection occurring between 100–200 ms post-stimulus onset, with its peak amplitude and latency quantified within this time window. Similarly, for the P3 component, stimulus evaluation associated with the task was captured by identifying and analyzing the largest positive peak within the 300–600 ms post-stimulus time frame.

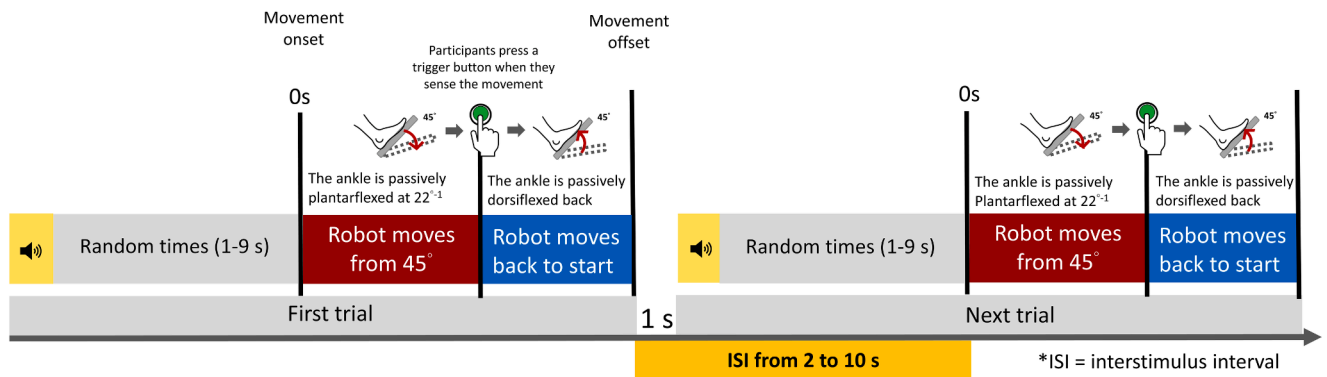
2.5. Statistical analysis

To investigate the difference in behavioral measures for detecting ankle motion between young adults with pDCD and controls, independent sample t-tests were performed for two behavioral measures: MDT_{mean} and MDT_{SD}, as the values met the criteria of normality and homogeneity of variance confirmed by Kolmogorov-Smirnov and Levene's tests. When a statistical difference was found, the effect size (Cohen's *d*) was reported to reflect the magnitude of the differences between groups. The significance level was set to $\alpha < 0.05$.

ERP measures were analyzed using mixed-design analyses of variance (ANOVAs), with the between-subject factor *GROUP* (pDCD vs. control) and within-subject factors *CONDITION* (VR, NVR) and *ELECTRODE* (Fz, FCz, Cz). A three-way ANOVA [2 (*GROUP*) $\times$ 2 (*CONDITION*) $\times$ 3 (*ELECTRODE*)] was performed for N1 and P3 components. The assumptions of homogeneity and normality of variance were verified using the Levene's and Kolmogorov-Smirnov tests, respectively. If violations of the sphericity assumption were identified in repeated-measures ANOVA, the Greenhouse-Geisser correction was applied. Post hoc comparisons between means were conducted using the Bonferroni test with a significance criterion of $\alpha < 0.05$. Additionally, the effect size (η_p^2) was reported.

Finally, to examine the relationship between behavioral and ERP measures, Spearman's rank correlation coefficients were calculated between behavioral measures (MDT_{mean} and MDT_{SD}) and N1 and P3 indices for each group separately. The significance level for the correlations was set to $\alpha < 0.05$.

A. Voluntary response (VR) condition



B. Non-voluntary response (NVR) condition

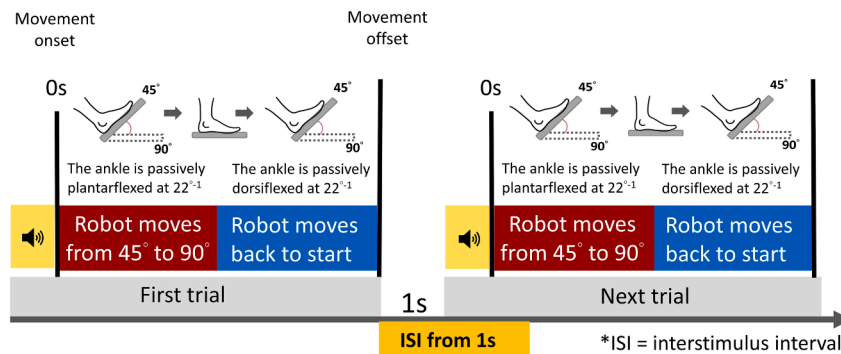


Fig. 2. Schematic diagram of the two proprioceptive task conditions. (A) Voluntary Response (VR) condition and (B) Non-Voluntary Response (NVR) condition.

3. Results

3.1. Behavioral performances: MDT_{mean} and MDT_{SD}

We evaluated MDT_{mean} and MDT_{SD} in the VR condition to understand how fast participants respond to ankle motion and the variability of responses within each group. The independent t -test analysis yielded a significant group difference ($t(22) = 5.03$, $p < 0.001$, Cohen's $d = 2.05$) for MDT_{mean} , indicating that young adults with pDCD ($MDT_{mean} M \pm SD$: 357.76 ± 4.88 ms) exhibited significantly longer mean ankle motion detection time when compared to the control group ($MDT_{mean} M \pm SD$: 277.78 ± 2.55 ms). Similarly, MDT_{SD} also revealed a significant group effect ($t(22) = 3.83$, $p < 0.001$, Cohen's $d = 1.56$), with the pDCD group ($MDT_{SD} M \pm SD$: 5.63 ± 2.08 ms) showing greater response variability than the control group ($MDT_{SD} M \pm SD$: 3.14 ± 0.89 ms) (see Fig. 3). According to Cohen's guidelines, these d values (greater than 0.8) indicate large effect sizes, suggesting substantial differences between the groups.

3.2. ERP indices

The ERP data for both N1 and P3 amplitude and latency met the criteria of normality and homogeneity of variance. The grand-average ERP waveforms from the Fz, FCz and Cz electrodes in the VR condition in the two groups are illustrated in Fig. 4A.

3.2.1. N1 component

We examined N1 amplitude to assess the cortical processing of proprioceptive input following the onset of ankle movement in two groups

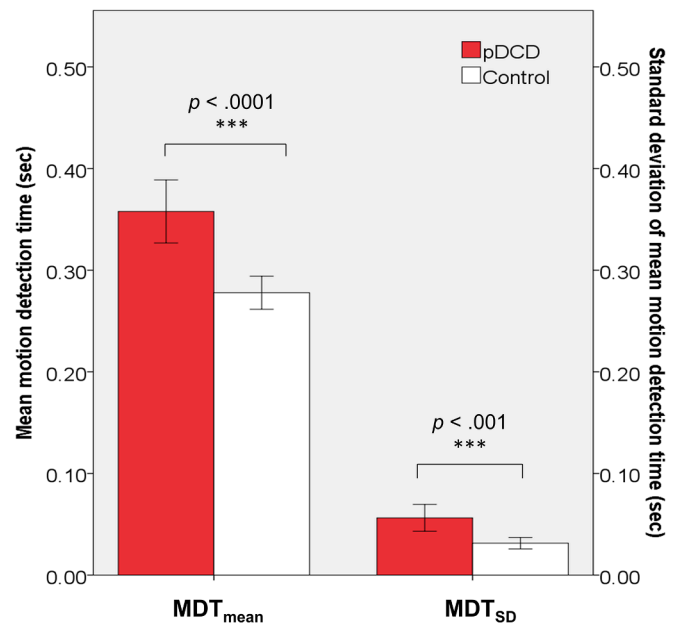


Fig. 3. Behavioral performance in the motion detection task. Bar graphs illustrating (A) mean motion detection time (MDT_{mean}) and (B) standard deviation of motion detection time (MDT_{SD}) for pDCD and control groups. Error bars represent standard deviation.

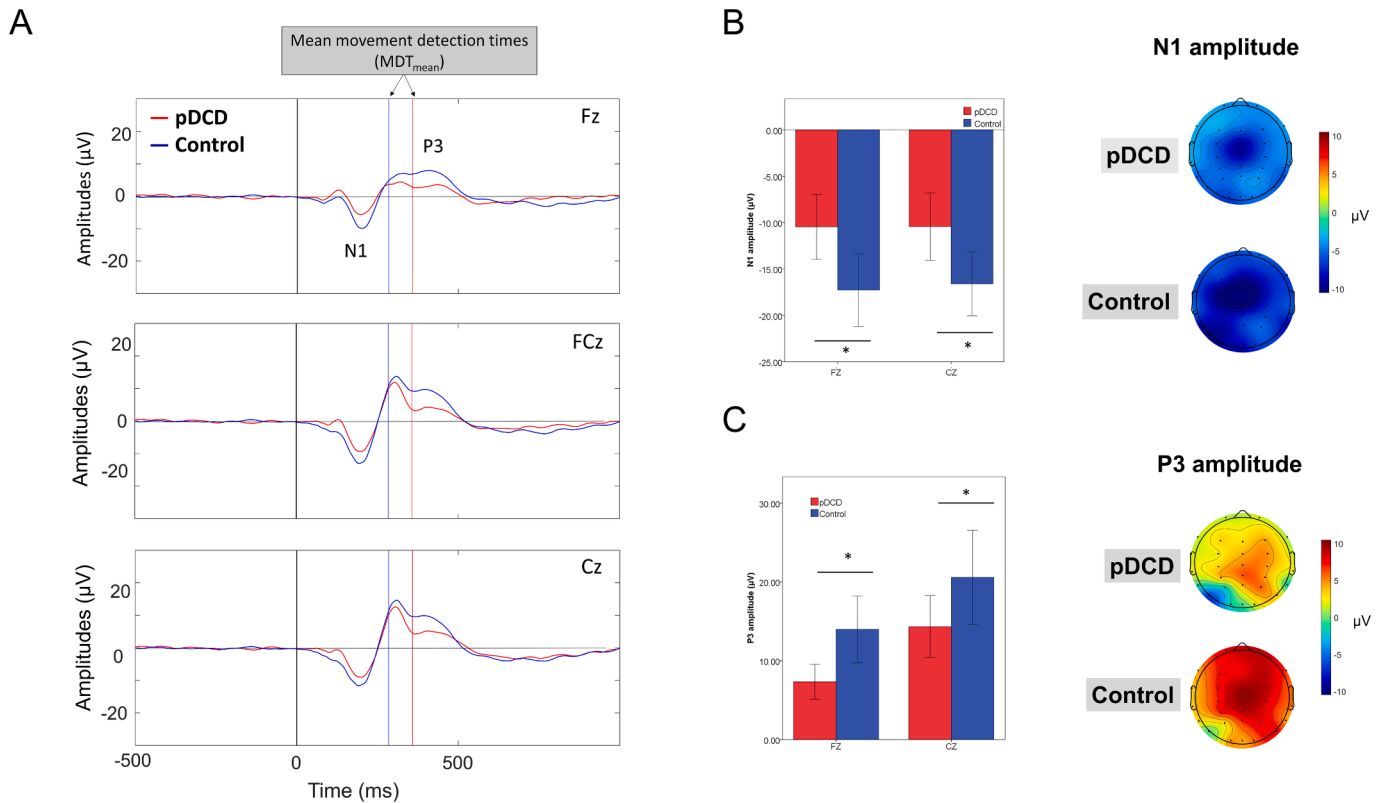


Fig. 4. ERP components during proprioceptive task. (A) Grand-average ERP waveforms from Fz, FCz, and Cz electrodes in the VR condition for pDCD and control groups. (B) N1 amplitude comparison between groups at Fz and Cz electrodes. (C) P3 amplitude comparison between groups at Fz and Cz electrodes. (Color figure available online.)

(pDCD and control), two conditions (VR and NVR), and three electrodes (Fz, FCz, and Cz). N1 amplitude reflects the early sensory processing of proprioceptive information. The three-way ANOVA revealed a significant main effect of *GROUP* [$F = 6.40, p = 0.019, \eta_p^2 = 0.23$], indicating that the pDCD group ($M \pm SD: -8.69 \pm 3.71 \mu V$) had overall smaller N1 amplitude when receiving the ankle motion stimuli compared to the control group ($M \pm SD: -12.53 \pm 3.72 \mu V$) (see Fig. 4B). It also yielded a significant main effect for *CONDITION* [$F = 7.47, p = 0.012, \eta_p^2 = 0.25$], indicating that N1 amplitude of the VR condition ($M \pm SD: -12.15 \pm 3.97 \mu V$) was larger than that of the NVR condition ($M \pm SD: -9.07 \pm 3.53 \mu V$) in all participants. Finally, a significant main effect of *ELECTRODE* was found for the N1 amplitude [$F = 55.92, p < 0.001, \eta_p^2 = 0.72$], showing that the N1 amplitude was largest at FCz ($M \pm SD: -12.20 \pm 4.29 \mu V$), followed by Cz ($M \pm SD: -11.76 \pm 4.18 \mu V$) and Fz ($M \pm SD: -7.90 \pm 3.24 \mu V$). The rest of the three-way and two-way interactions were not significant (p 's > 0.05).

N1 latency was examined to investigate the timing of early cortical processing in response to proprioceptive stimuli following the onset of ankle movement. The results showed a significant main effect for *CONDITION* [$F = 46.06, p < 0.001, \eta_p^2 = 0.68$], indicating that N1 latency of the VR condition ($M \pm SD: 183.33 \pm 28.79$ ms) was longer than that of the NVR condition ($M \pm SD: 115.00 \pm 30.16$ ms). The remaining main effects and interaction effects were not significant (p 's > 0.05).

3.2.2. P3 component

P3 amplitude was examined to understand the stimulus evaluation (i. e., motor responses) corresponding to the ankle proprioceptive stimuli. The ANOVA revealed a significant main effect for *GROUP* [$F = 5.22, p = 0.032, \eta_p^2 = 0.19$], indicating that young adults in the pDCD group ($M \pm SD: 9.54 \pm 4.20 \mu V$) exhibited smaller P3 amplitude when responding to the ankle motion stimuli compared to the controls ($M \pm SD: 13.46 \pm$

$6.54 \mu V$) (see Fig. 4C). Another significant main effect was found for *CONDITION*, showing that P3 amplitude in the VR condition ($M \pm SD: 15.06 \pm 4.20 \mu V$) was greater than that of the NVR condition ($M \pm SD: 7.95 \pm 3.86 \mu V$). Notably, a significant interaction effect between *GROUP* and *CONDITION* was also detected.

To further investigate the interaction effect, we assessed the group differences in each condition separately. In the VR condition, young adults with pDCD exhibited a significantly smaller P3 amplitude ($M \pm SD: 9.54 \pm 4.74 \mu V$) compared to the control group ($M \pm SD: 13.46 \pm 6.24 \mu V$). However, no group difference was found in the NVR condition and the rest of the main effects and interaction effects (p 's > 0.05).

For the P3 latency, the three-way ANOVA analysis revealed non-significant main effects as well as interaction effects (p 's > 0.05). These results suggest that P3 latency was not influenced by group (pDCD vs. control), task condition (VR vs. NVR), or electrode site (Fz, FCz, Cz), either independently or jointly.

3.3. Correlation between behavioral performances and ERP measures

We investigated how N1 and P3 components correlated with 1) behavioral performances (MDT_{mean} , MDT_{SD}) and 2) different domains of BOT-2 scores [fine motor control (FMC), manual coordination (MC), body coordination (BC), and strength and agility (SA)] in the complete sample of young adults. To reduce the type I error, the ERP outcomes across three electrodes (Fz, FCz, Cz) were averaged for the following correlation analysis.

For the complete sample, N1 amplitude in the VR condition showed significant correlation with all behavioral measures. Stronger N1 amplitude was associated with faster and more stable responses to ankle motion ($MDT_{mean}: r_s = 0.62, p < 0.001$; $MDT_{SD}: r_s = 0.51, p = 0.011$) (see Fig. 5A). Additionally, stronger N1 amplitude was linked to higher levels of motor skills, demonstrated by negative correlations with MC ($r_s =$

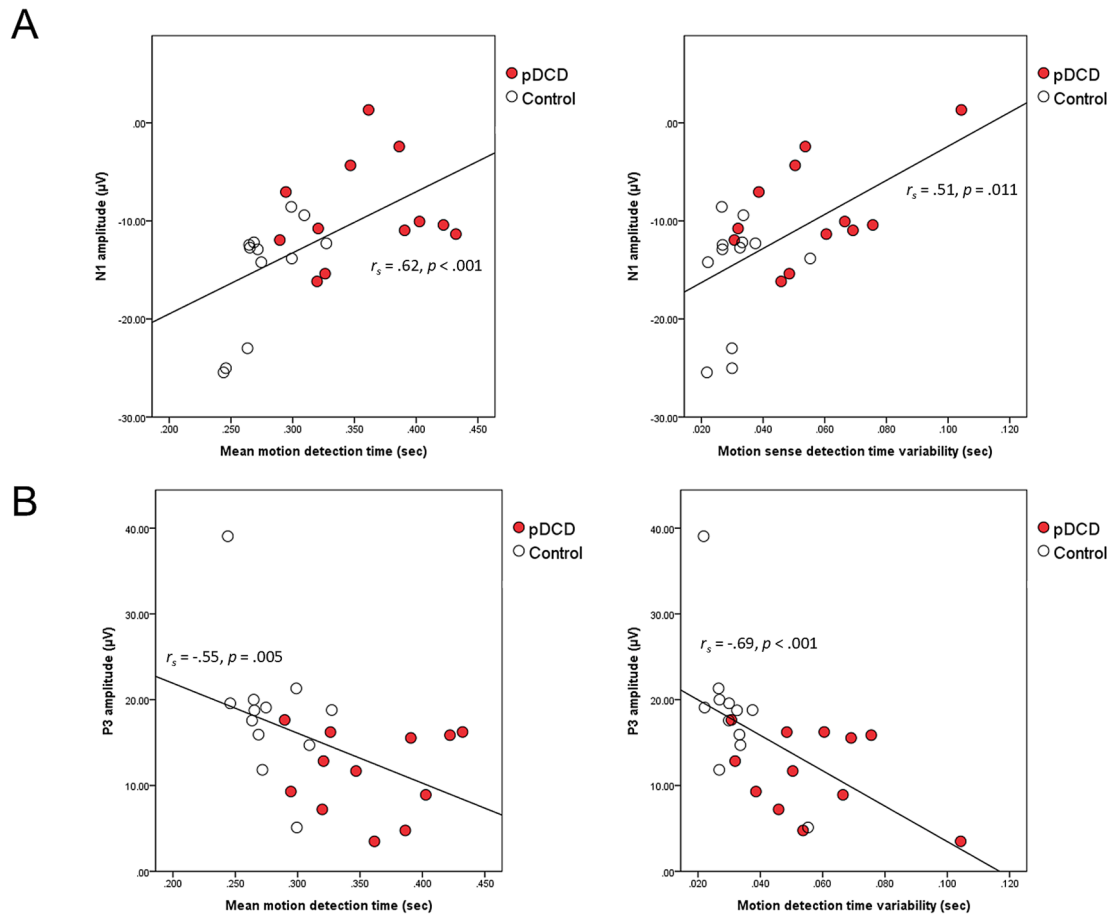


Fig. 5. Correlations between behavioral measures and neuroelectric indices in the VR condition. Scatter plots depicting **A)** significant negative correlations between N1 amplitude (averaged across Cz, Fz, and FCz electrodes) and both mean motion detection time (MDT_{mean}) and standard deviation of motion detection time (MDT_{SD}) and **B)** significant positive correlations between P3 amplitude and both MDT_{mean} and MDT_{SD} . Individuals with smaller N1 and P3 amplitudes exhibited longer ankle motion detection times and larger response variability. Data points represent individual participants from both pDCD and control groups.

-0.50; $p = 0.013$), BC ($r_s = -0.46$, $p = 0.023$), and SA ($r_s = -0.48$, $p = 0.019$) (see Fig. 6). Similarly, P3 amplitude in the VR condition was related to four behavioral outcomes, indicating that higher P3 amplitude was associated with shorter MDT_{mean} ($r_s = -0.55$, $p = 0.005$), smaller MDT_{SD} ($r_s = -0.69$, $p < 0.001$), and higher levels of MC ($r_s = 0.60$, $p = 0.002$) and BC ($r_s = 0.59$, $p = 0.003$) on the standardized BOT-2 test (see Fig. 5B). No significant relationship was found between the N1/P3 latency and the behavioral measures.

In the control group, stronger N1 amplitude in the VR condition

correlated with faster motion detection time (MDT_{mean} : $r = 0.66$, $p = 0.018$). Moreover, N1 amplitude in the NVR condition also linked to BOT-2 MC ($r = -0.64$, $p = 0.025$) and total motor scores ($r = -0.73$, $p = 0.007$), suggesting that stronger N1 amplitude was associated with higher levels of motor skills. In the P3 component, stronger P3 amplitude in the VR condition was linked to MDT_{SD} ($r_s = -0.64$, $p = 0.024$), and faster latency correlated with higher total motor scores ($r_s = -0.60$, $p = 0.040$). In the NVR condition, stronger P3 amplitude was again associated with smaller MDT_{SD} ($r_s = -0.59$, $p = 0.045$) and higher total

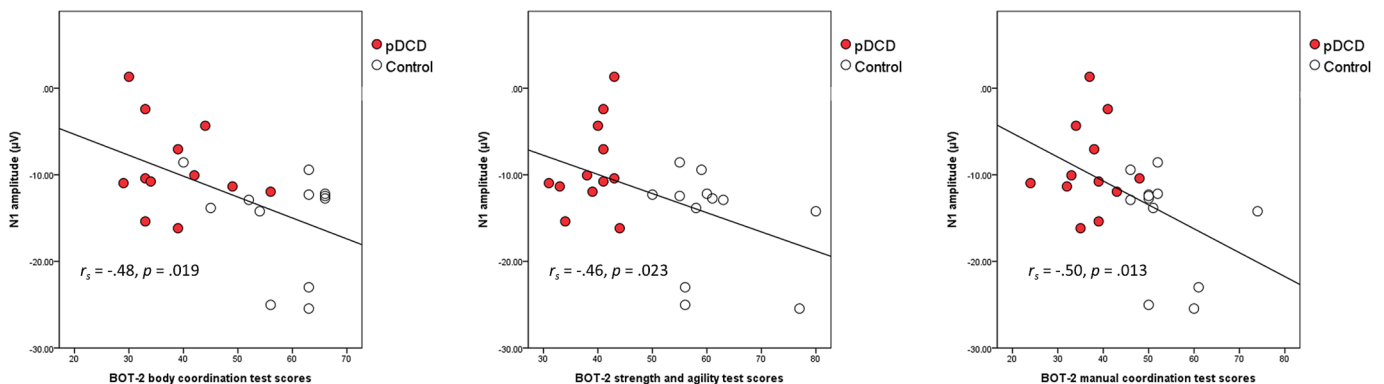


Fig. 6. Correlations between neuroelectric indices and clinical BOT-2 test scores. Scatter plots illustrating significant negative correlations between N1 amplitude (averaged across Cz, Fz, and FCz electrodes) and BOT-2 subscores for body coordination, strength and agility, and manual coordination. Individuals with smaller N1 amplitudes demonstrated poorer motor performances. Data points represent individual participants from both pDCD and control groups.

motor scores ($r_s = 0.67, p = 0.017$), while shorter P3 latency correlated with smaller MDT_{SD} ($r_s = 0.65, p = 0.022$). These findings indicate that healthy young adults with stronger N1 and P3 amplitude along with faster latency, consistently demonstrated better performance on proprioceptive tasks and exhibited higher levels of motor skills.

For the pDCD group, the results yielded no significant correlation between behavioral performances and ERP measures (p 's > 0.05).

4. Discussion

This is the first study to examine the behavioral and cortical brain indices associated with proprioceptive-motor processing in response to ankle motion in young adults with and without pDCD. Specifically, we compared behavioral performances (i.e., mean movement detection time and the standard deviation of the movement detection time) and ERP indices (i.e., N1 and P3 amplitude and latency) between young adults with and without pDCD during VR and NVR conditions to passive ankle motion. The results showed that individuals with pDCD exhibited slower and more variable responses in sensing ankle motion. The ERP data revealed reduced N1 amplitude following the proprioceptive stimulus, and smaller P3 amplitude during stimulus evaluation in response to corresponding ankle motion. Furthermore, significant correlations were found between ERP measures (N1 and P3 amplitudes) and behavioral performances (MDT_{mean} and MDT_{SD}) as well as motor skill levels (BOT-2 domain scores) in the overall sample and the control group, but not in the pDCD group. To the best of our knowledge, these findings are the first to provide novel insights into the neural underpinnings of proprioceptive-motor deficits in young adults with pDCD. We will discuss the main findings as follows.

4.1. Behavioral measures: Individuals with pDCD exhibited slower ankle motion detection times

Our data showing slower and more variable responses to ankle motion in young adults with pDCD are consistent with previous research demonstrating lower limb proprioceptive deficits in individuals with DCD (Chen et al., 2020; Tseng et al., 2022; Tseng et al., 2018; Tseng et al., 2023). In our study, the time to detect passive ankle motion is significantly longer in the pDCD group, shown by increased MDT_{mean} values of 28.8 % when compared to their controls. This result is aligned with recent work by Tseng et al. (2023), reported a reduction of 22–59 % in ankle motion detection time across different velocities of ankle plantarflexion in children with pDCD. Moreover, we observed that MDT_{SD} , the variability among the repetitive responses, was also altered in the pDCD group. This finding is consistent with the increased variability in proprioceptive responses reported in previous studies (Chen et al., 2020; Tseng et al., 2018, 2019). The increased MDT_{mean} and MDT_{SD} in the pDCD group suggest that proprioceptive-motor function in the lower limbs is impaired in young adults with pDCD.

Currently, there is no single gold standard method to measure lower limb motion sense. While our findings align with those of previous studies, it is noteworthy to acknowledge some methodological differences among studies. For example, Tseng et al. (2023) employed nine different velocities (0.15°/s to 1.35°/s) and calculated the adjusted motion detection time to measure proprioceptive sensitivity by subtracting the simple reaction time from the total movement detection time. In contrast, our study utilized a fixed angular velocity of 22°/s for ankle plantarflexion to measure proprioceptive-motor function, including the time from the onset of movement to the offset of the motor response. The use of a fixed velocity allowed us to eliminate the potential confounding effects of velocity on proprioceptive-motor function and enabled simultaneous EEG recording during the task, providing insights into the neural correlates of proprioceptive processing in individuals with pDCD. Despite these methodological differences, both studies highlight the importance of assessing proprioceptive function in individuals with pDCD. Future studies should consider incorporating

both proprioceptive sensitivity (i.e., different motion velocities) and the subsequent proprioceptive-motor function measures to offer a more comprehensive assessment of proprioceptive processing in individuals with pDCD.

4.2. ERP measures: Reduced N1 and P3 amplitude after the proprioceptive stimulus in individuals with DCD

Our study addresses a critical gap in understanding the neural mechanisms underlying proprioceptive deficits in pDCD using ERPs. Specifically, we investigated the cortical processing of ankle proprioceptive information, a key component in maintaining balance and coordinating lower limb movements. The findings reveal significantly reduced N1 and P3 amplitudes in response to passive ankle motion stimuli in the pDCD group compared to controls, across both proprioceptive conditions. This reduction indicates robust impairments in proprioceptive information processing in the pDCD group.

The N1 component, primarily generated in the primary and secondary somatosensory cortices, showed diminished amplitude in the pDCD group across both task conditions (VR and NVR). The reduction in N1 amplitude of the central and frontal regions suggests a deficit in the initial detection and encoding of proprioceptive input that extends beyond early sensory processing to areas involved in motor planning and execution (Mikami et al., 2021; Tran et al., 2022). This attenuated N1 response implies that individuals with pDCD may struggle with the primary stages of processing proprioceptive information from their muscles and joints, indicating a disruption in the proprioceptive pathway from peripheral receptors to cortical brain levels (Toledo et al., 2016; Toledo, Manzano, et al., 2016). The observed deficits in early processing (N1) appear to cascade into later stages of information processing, as evidenced by the reduced P3 amplitude in the pDCD group, particularly in the condition with voluntary movements. The P3 component, believed to originate from multiple cortical sources including the parietal lobe, frontal cortex, and temporoparietal junction, was also diminished. This component is associated with higher-order cognitive processes such as stimulus evaluation and response selection (Kida et al., 2003; Polich, 2007). The reduction in P3 amplitude across our targeted electrode sites indicates that the initial deficits in sensory processing extend to more complex stages of information processing, affecting areas involved in executive function, motor planning and execution (Tsai et al., 2009; Tsai et al., 2012). This sequential impairment, from reduced N1 to diminished P3, suggests that individuals with pDCD face challenges in both the early detection of proprioceptive stimuli and the subsequent cognitive processing of this information.

Interestingly, despite the significant reductions in N1 and P3 amplitudes, there were no significant differences in the latencies of these components between the pDCD and control groups. This finding suggests that the timing of neural processes remains intact, while the strength of neural activation is compromised. Thus, it appears that neural activation strength, rather than timing, may be the primary factor contributing to proprioceptive processing deficits in young adults with pDCD. Taken together, these ERP findings provide a neurophysiological basis for understanding the proprioceptive deficits observed in DCD. They highlight a continuum of impairment from early sensory detection to higher-order processing, which may contribute to the motor coordination difficulties characteristic of this disorder.

4.3. The correlation between ERP and behavioral data

We found significant correlations between ERP components (N1 and P3) and behavioral outcomes in the complete sample and control group. In the VR condition, larger N1 and P3 amplitudes were associated with faster (MDT_{mean}) and more consistent responses (MDT_{SD}) to ankle motion in the entire sample and the control group. These findings suggest that stronger neural activations during early sensory processing and stimulus evaluation stages contribute to better performance in detecting

proprioceptive stimuli. The relationship between N1 amplitude and MDT_{mean}/MDT_{SD} indicates that efficient early processing of proprioceptive information is crucial for rapid and consistent detection of passive ankle movements. Similarly, the correlation between P3 amplitude and MDT_{mean}/MDT_{SD} implies that the allocation of attentional resources for stimulus evaluation and response preparation facilitates faster and more stable responses to proprioceptive stimuli.

Furthermore, we observed significant correlations between these ERP components and BOT-2 scores in the complete sample, with larger N1 and P3 amplitudes in the VR condition associated with better manual coordination, body coordination, and strength and agility. These findings suggest that neural activations during proprioceptive processing are related to both task-specific performance and overall motor competence, highlighting the interconnectedness of somatosensory processing, cognitive evaluation and motor skills. Our findings are aligned with previous studies reporting associations between ERP components and behavioral performance in cognitive and sensorimotor tasks (Kao et al., 2022; Ludyga et al., 2021), and extend these findings to standardized measures of motor proficiency. Collectively, these observations highlight the importance of efficient proprioceptive processing and its associated cognitive evaluation in supporting proficient motor control and coordination, while also suggesting that ERPs could serve as valuable objective markers for assessing motor function and motor development in this population.

It is important to note that none of the correlations in the pDCD group were significant. This lack of significant correlations may be attributed to the heterogeneous nature of DCD, characterized by a diverse spectrum of sensory and motor challenges (Tseng et al., 2022). For instance, some individuals with DCD predominantly exhibit visual processing deficits, while others display more pronounced proprioceptive impairments. To address these limitations, future research should focus on characterizing specific profiles of individuals with pDCD and comprehensively examining proprioceptive function in both upper and lower extremities. Moreover, investigating the associations between site-specific proprioceptive deficits and motor control measures could yield a more nuanced understanding of DCD. These approaches may elucidate the complex interaction between neural processing, proprioception, and motor function in DCD, ultimately informing the development of more targeted and effective interventions.

5. Conclusions

This study reveals novel neural correlates of proprioceptive-motor deficits in young adults with pDCD. Reduced N1 and P3 amplitudes in ERPs during a proprioceptive-motor task suggest impairments in early sensory processing and higher-order cortical processing of proprioceptive inputs, which correlate with behavioral indices and motor skills competence. These findings implicate deficient cortical processing of proprioceptive signals as a contributor to the movement challenges observed in pDCD. Further research characterizing the heterogeneity within the pDCD population and the developmental trajectory of proprioceptive-motorfunction, along with multi-modal assessments, may elucidate brain-behavior relationships and inform targeted sensorimotor interventions for individuals with pDCD.

CRedit authorship contribution statement

Yu-Ting Tseng: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Writing – original draft, Writing – review & editing. **Yi-Wen Chen:** Data curation, Formal analysis, Writing – review & editing. **Yi-Hsuan Lin:** Data curation, Formal analysis, Writing – review & editing. **Chia-Liang Tsai:** Resources, Software, Writing – review & editing. **Chun-Hsiang Chuang:** Data curation, Resources, Software, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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